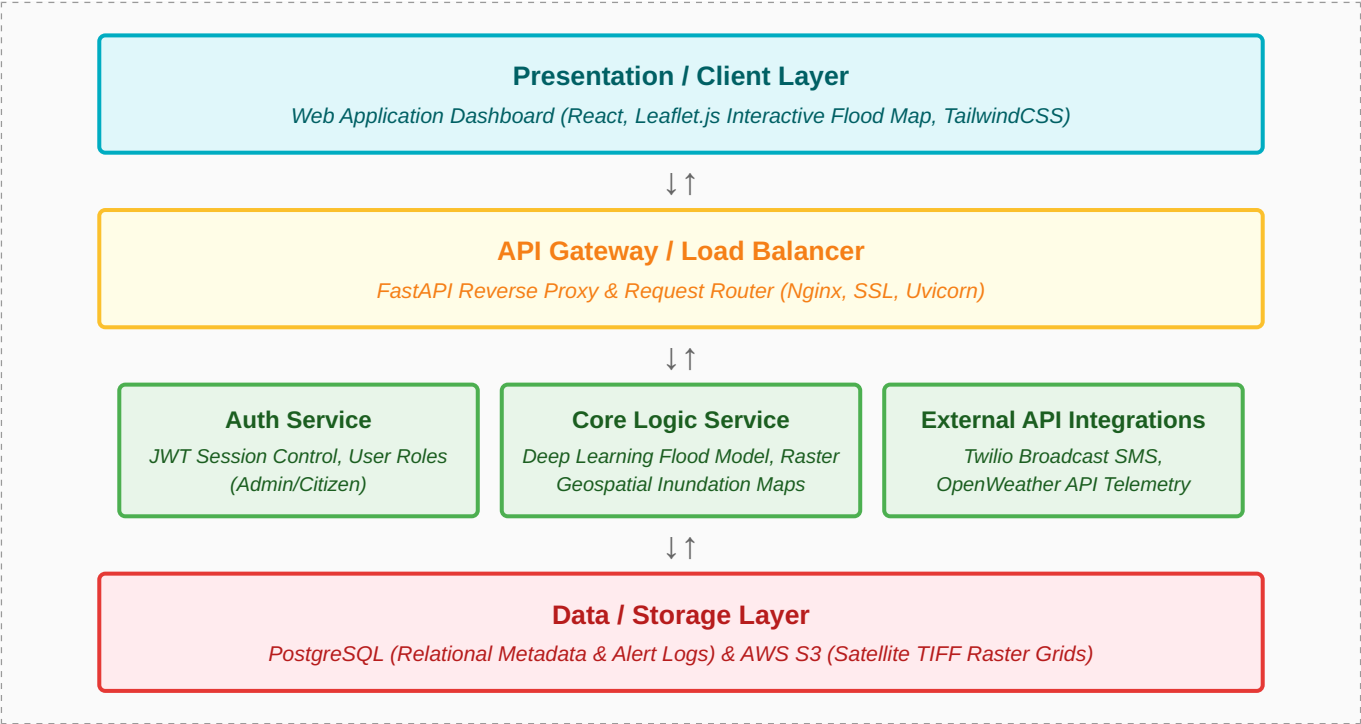


Solution Architecture

Date	15 July 2026
Team ID	TM-RISINGWATERS-04
Project Name	Rising Waters: AI Powered Flood Prediction System
Maximum Marks	5 Marks

Solution Architecture Diagram:

The high-level visual representation of the Solution Architecture maps out the dynamic Presentation Layer, the robust Application Services routing core, and the integrated geospatial and telemetry storage systems designed for real-time inundation modeling.



Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides the responsive graphical interface for municipal rescue coordinators and the public. Displays color-coded real-time street level predictive maps, critical risk statistics, and diagnostic hazard levels.	React, Leaflet.js, TailwindCSS, Chart.js
API Gateway	Serves as the secure ingress point for all client dashboard queries. Handles request throttling, routes traffic to respective background workers, and enforces authentication handshakes.	Nginx, FastAPI (Uvicorn), SSL Encryption
Microservices / Application Services	Core Flood Model Service: Executes heavy computational forecasting using the trained deep learning algorithm. Notification Dispatcher: Interacts with cellular networks to broadcast automated evacuation alerts to localized grid contacts. Session Orchestrator: Handles user signup, administrative settings, and JWT verification.	PyTorch, Rasterio, Celery (Async Tasks), Redis queue, Twilio API, JWT
Database / Storage Layer	Stores geographical bounds coordinates, historical sensor stream telemetry database records, and critical subscriber contact lists. Handles multi-band raster digital elevation models (.tif format).	PostgreSQL (with PostGIS extensions), AWS S3 bucket storage